

Infrared Interferometric Estimation of Silicon Carbide Epitaxial-Layer Thickness: Cross-Angle and Cross-Model Consistency

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Abstract—Fourier-transform infrared (FTIR) reflectance fringes can be used to estimate epitaxial-layer thickness without destructive sectioning, but the result depends on the refractive-index model, incidence geometry, fringe localization, and the treatment of repeated internal reflections. We examine those dependencies using four reflectance spectra released with the 2025 China Undergraduate Mathematical Contest in Modeling (CUMCM) Problem B. For the two SiC spectra measured at 10° and 15° , a dispersive two-beam estimator gives thicknesses of $8.50\text{ }\mu\text{m}$ and $8.60\text{ }\mu\text{m}$; an Airy-type multiple-reflection treatment gives $8.55\text{ }\mu\text{m}$ and $8.63\text{ }\mu\text{m}$. The changes between optical models are 0.59% and 0.35% , and the two-beam estimates differ by 1.18% across angle. Two silicon spectra analyzed with the multiple-reflection model give $3.79\text{ }\mu\text{m}$ and $3.53\text{ }\mu\text{m}$. Analytical sensitivity coefficients show that refractive-index and fringe-spacing errors dominate small incidence-angle errors at the present geometry. The released data do not include an independently calibrated reference thickness or a complete instrument uncertainty budget; accordingly, the reported agreement is interpreted as internal consistency and model stability, not traceable absolute accuracy.

Index Terms—FTIR reflectance, silicon carbide, epitaxial layer, thin-film interference, thickness measurement, uncertainty.

I. INTRODUCTION

SILICON carbide (SiC) epitaxial layers are used in power and high-temperature electronic devices, where layer thickness is an important process-control quantity. FTIR reflectance offers a non-destructive route to thickness estimation because the reflections from the air/layer and layer/substrate interfaces interfere. The spectral period carries optical-thickness information, but converting that period to geometric thickness is not a purely geometric exercise. Dispersion, incidence angle, absorption, polarization, fringe localization, and internal round trips can all move the estimate.

The basic fringe-spacing idea is well established in thin-film metrology. Swanepoel related interference extrema to film thickness and optical constants [7], while standard thin-film treatments show how phase and interface reflectance enter multilayer response [6]. For SiC, Camassel *et al.* demonstrated that IR reflectance is sensitive to the structure of doped SiC stacks [1]; Tang *et al.* explicitly incorporated the frequency dependence of refractive index into 4H-SiC thickness estimation [2]; and later frequency-domain and phase-resolved analyses have treated multilayer structure, dispersion, absorption, and phase in greater detail [3], [4].

The measurement question here is narrower. We ask how much a transparent fringe-spacing estimate changes when two practical modeling choices are altered: the external incidence angle and the treatment of repeated internal reflection. That question can be answered with the released spectra even though the dataset does not provide a certified reference thickness. It cannot support a traceable absolute-accuracy claim, and we do not make one.

This distinction matters because a high correlation or a small model-to-model difference is not the same thing as measurement accuracy. A fit statistic describes agreement within a chosen model; metrological uncertainty requires an explicit measurement equation and defensible uncertainty assigned to its inputs. We therefore report three separate checks: cross-angle agreement, cross-model agreement, and local sensitivity of the estimator to fringe spacing, refractive index, and angle. The uncertainty discussion follows the logic of the Guide to the Expression of Uncertainty in Measurement (GUM) rather than converting fit quality into a percentage accuracy [5].

The spectra were supplied as part of CUMCM 2025 Problem B; the authors did not perform the physical measurements. The contribution is consequently methodological: a reproducible derivation of a dispersive two-beam estimator, an Airy-type cross-check for repeated reflections, a polarization check at the larger incidence angle, and a measurement-oriented interpretation of the resulting consistency and limitations.

II. DATA AND OPTICAL MODEL

A. Reflectance spectra and analysis window

The dataset contains two SiC reflectance spectra measured at external incidence angles $\theta_0 = 10^\circ$ and 15° , together with two silicon-wafer spectra. Figure 1 shows the two SiC spectra. A pronounced disturbance occurs around the SiC lattice-vibration region near $800\text{--}1000\text{ cm}^{-1}$, where strong absorption and rapidly varying optical constants distort the regular interference pattern. Thickness estimation is therefore restricted to $2500\text{--}4000\text{ cm}^{-1}$, where the fringes are substantially more regular.

The refractive index is modeled as dispersive using the Sellmeier-type relation employed in the analysis,

$$n^2 = 1 + \frac{0.20075\lambda^2}{\lambda^2 + 12.07224} + \frac{5.54861\lambda^2}{\lambda^2 - 0.02641} + \frac{35.65066\lambda^2}{\lambda^2 - 1268.24708}, \quad (1)$$

with the wavelength convention used in the supplied calculation. Within the selected band, neighboring extrema are separated

Reflectance vs. Wavenumber at Two Incidence Angles

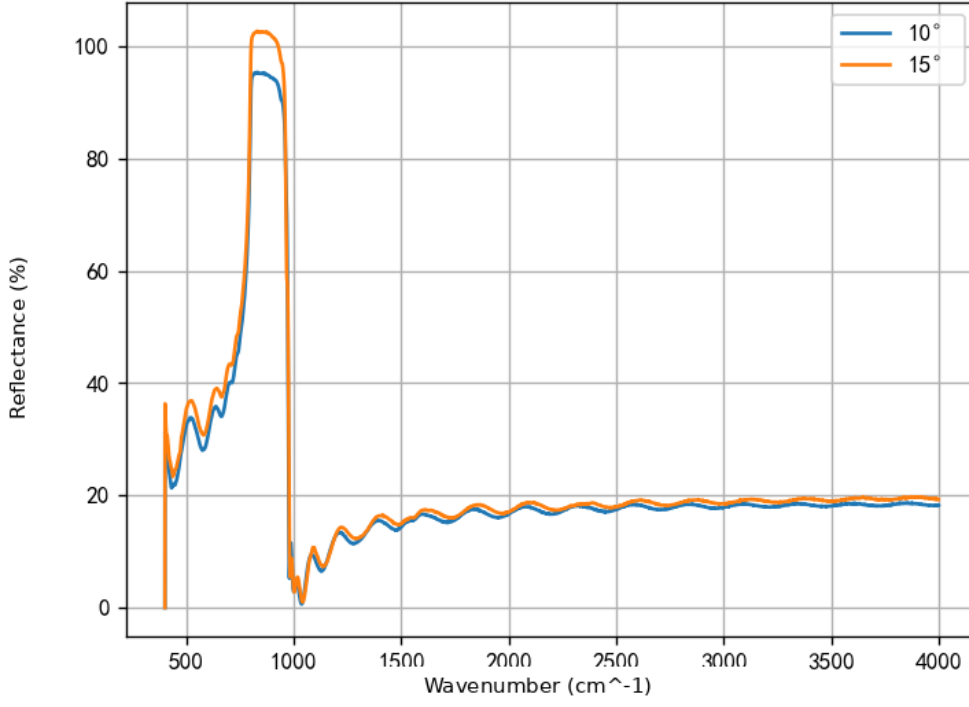


Fig. 1. Reflectance spectra at the two SiC incidence angles. The thickness analysis uses the high-wavenumber region in which regular fringes remain identifiable.

by roughly $120\text{--}135\text{ cm}^{-1}$; the corresponding local change in the calculated refractive index is approximately 1% or less. We therefore retain dispersion across the full analysis band while treating n as locally constant over one fringe interval.

B. Two-beam model

Let d be geometric layer thickness, n the local refractive index, θ_0 the external angle, and θ_t the refracted angle given by Snell's law. Retaining the direct surface reflection and one internally reflected component gives an approximately sinusoidal interference term. In wavenumber $s = 1/\lambda$, the phase grows linearly with sd through the angular optical-path factor

$$G(n, \theta_0) = \frac{2n^2 - \sin^2 \theta_0}{2\sqrt{n^2 - \sin^2 \theta_0}}. \quad (2)$$

For adjacent reliable extrema separated by Δs , the local thickness estimator used here is

$$d = \frac{1}{4\Delta s} G(n, \theta_0)^{-1}. \quad (3)$$

The extrema envelope is also used to infer the local interface reflectance and hence n under the scalar Fresnel approximation. Figure 2 illustrates the optical geometry.

C. Multiple-reflection model

When interface reflectances are appreciable, absorption is weak, and the interfaces preserve coherence, repeated round

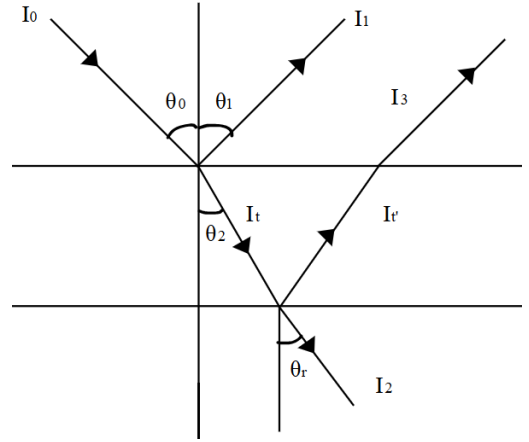


Fig. 2. Optical geometry for the reflected components used in the fringe-spacing derivation.

trips form a geometric series. With amplitude coefficients r_1, r_2, t_1, t_2 and round-trip phase

$$\delta = \frac{2\pi}{\lambda} 2nd \cos \theta_t, \quad (4)$$

the transmitted field is proportional to $(1 - r_1 r_2 e^{i\delta})^{-1}$. The corresponding normalized reflectance takes the familiar Airy-type form

$$R = 1 - \frac{(1 - R_1)(1 - R_2)}{1 + R_1 R_2 - 2\sqrt{R_1 R_2} \cos \delta}. \quad (5)$$

Extrema of the transformed quantity $1/(1-R)$ allow R_1R_2 and the two interface reflectances to be recovered before Eq. (3) is applied. The transformation therefore changes how the optical envelope is inferred while preserving the same phase-spacing dependence of thickness.

D. Model hierarchy and estimator positioning

The two optical treatments used here occupy different points in a hierarchy of reflectance models. The two-beam formulation is intentionally transparent: the observed period is mapped to thickness through a local refractive index and an angular correction. Its main advantage is auditability, because errors in fringe spacing, refractive index, or incidence geometry have a direct interpretation. Its limitation is that the amplitude envelope and repeated reflections are simplified.

The Airy-type formulation retains the coherent infinite series of internal round trips and therefore provides a direct check on whether the simplified envelope treatment changes the recovered thickness. It is still not a full dispersive inversion: absorption is controlled primarily by spectral-window selection, and the local optical constants are inferred from extrema rather than fitted simultaneously over the entire spectrum. A full transfer-matrix or phase-resolved fit would occupy the next level of the hierarchy by treating complex optical constants, background, and phase over the complete usable band [6], [4].

This hierarchy motivates the interpretation used in this paper. Agreement between the two-beam and Airy-type estimates is evidence that repeated reflection is a small model correction for the two SiC spectra under the selected window; it is not evidence that all optical-model error is negligible. The comparison therefore functions as a model-discrepancy diagnostic rather than as an independent calibration.

III. ESTIMATION PROCEDURE AND MEASUREMENT CHECKS

A. Extrema selection

Peak locations are sensitive to local spectral noise. The procedure anchors the first and last reliable extrema inside the selected high-wavenumber window and uses the approximately regular fringe spacing to stabilize intermediate phase-order assignments. Local refractive indices and thicknesses are then computed from adjacent extrema and aggregated. This regularization reduces sensitivity to a single displaced local maximum but introduces a model-selection component: incorrect fringe order or peak placement directly perturbs Δs and therefore the recovered d .

B. Polarization check

The scalar Fresnel approximation is tested at the larger incidence angle. Taking $\theta_0 = 15^\circ$ and a representative $n \approx 2.5$ gives $R_s \approx 19.41\%$ and $R_p \approx 17.35\%$, whereas the unpolarized average is approximately 18.38%. The normal-incidence scalar approximation is approximately 18.37%. The average is therefore essentially unchanged for this geometry, supporting the neglect of polarization splitting in the subsequent thickness calculation. This conclusion is specific to the modest angles and index contrast considered here and should not be generalized to near-Brewster geometries.

C. Fit diagnostics

For the multiple-reflection treatment, the transformed spectra are approximately periodic in wavenumber, as illustrated in Figure 3. The reported correlations quantify linear/periodic fit consistency in the transformed analysis; they are not confidence levels and are not interpreted as percentage measurement accuracy.

D. Practical measurement sequence

For an implementation outside the competition dataset, the calculation can be organized as a short measurement chain. First, acquire reflectance with a recorded angle, spectral resolution, polarization state (or an explicitly unpolarized configuration), and instrument background. Second, remove spectral regions in which strong absorption or rapidly changing optical constants destroy the regular fringe pattern. Third, locate a sequence of extrema and record how peak position is determined; the spacing uncertainty should come from the spectral resolution and the peak-localization procedure rather than from a correlation coefficient. Fourth, evaluate the refractive index at the local wavenumbers and propagate any uncertainty in the dispersion relation. Finally, compare at least one simplified optical model with a model that retains additional physics, such as coherent internal reflections or a transfer-matrix description.

This sequence separates quantities that can be calibrated from choices that are model dependent. Angle, spectral scale, and repeatability can be characterized experimentally. Dispersion and absorption require material-specific optical information. Fringe-order selection and spectral-window selection are analysis choices and should be frozen before reporting the result. The cross-model comparison then serves as a check on model discrepancy instead of a substitute for calibration.

E. Conditions under which the estimator can fail

The estimator relies on an identifiable periodic phase in the usable spectrum. It becomes unreliable when absorption suppresses the fringes, when the layer is laterally nonuniform over the illuminated spot, when interfaces are rough enough to reduce coherence, when a fringe is missed or assigned the wrong order, or when the optical constants change too rapidly to be treated locally. Large incidence angles also make polarization splitting more important. Repeated reflections are not automatically negligible: they become more visible when interface reflectances are appreciable and absorption is low. These are not edge cases to be hidden in post-processing; they determine whether the measurement equation is an adequate representation of the experiment.

IV. RESULTS

Table I summarizes the recovered thicknesses. The two-beam SiC estimates are $8.50\mu\text{m}$ and $8.60\mu\text{m}$ at 10° and 15° , respectively. The corresponding multiple-reflection estimates are $8.55\mu\text{m}$ and $8.63\mu\text{m}$. Relative to the two-beam values, the changes are 0.59% and 0.35%, both small on the scale of the recovered thickness.

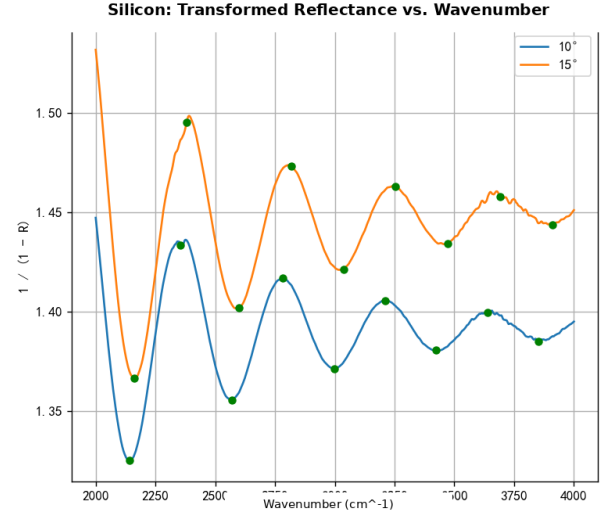
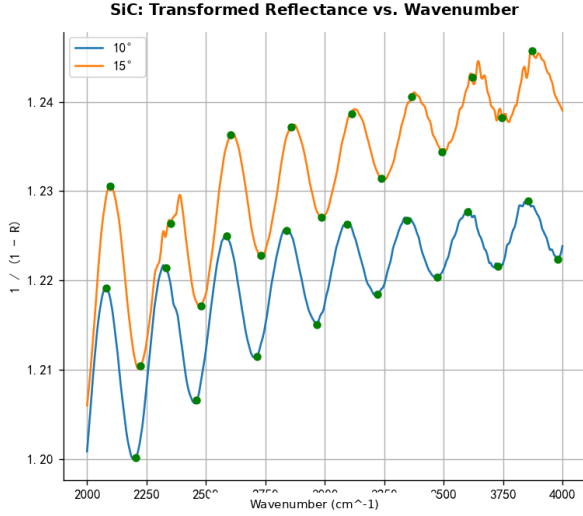


Fig. 3. Transformed spectra used in the multiple-reflection calculation. SiC cases are shown at left and silicon cases at right. The transformed periodicity is used as a fit-consistency diagnostic; it is not interpreted as a confidence level or measurement accuracy.

TABLE I
THICKNESS ESTIMATES AND UNAMBIGUOUS FIT DIAGNOSTICS RECORDED IN THE ANALYSIS.

Dataset	Optical treatment	Thickness (μm)	Fit diagnostic
SiC 1 (10°)	Two-beam	8.50	99.98%
SiC 2 (15°)	Two-beam	8.60	99.96%
SiC 1	Multiple reflection	8.55	99.91%
SiC 2	Multiple reflection	8.63	—
Si 3	Multiple reflection	3.79	99.97%
Si 4	Multiple reflection	3.53	99.96%

The retained calculation record contains two inconsistent values (99.96% and 99.83%) for the SiC-2 multiple-reflection fit diagnostic. Because this statistic is not needed to obtain the reported thickness and cannot be resolved from the available record, it is omitted rather than selecting one value.

The cross-angle difference between the two SiC two-beam results is $0.10\mu\text{m}$, about 1.18% relative to $8.50\mu\text{m}$. This provides an internal angular-consistency check. It should not be interpreted as a calibration uncertainty because the two spectra are not independent certified thickness standards.

The multiple-reflection calculation also yields mean recovered refractive indices of 2.47 and 2.53 for the two SiC spectra, with variances 5.79×10^{-4} and 1.18×10^{-4} , respectively. The silicon cases yield mean indices 3.25 and 3.49. These small within-window variances support the local-constant approximation over neighboring fringes but do not quantify systematic error in the refractive-index model.

V. UNCERTAINTY, ACCURACY, AND DISCUSSION

A measurement-oriented interpretation must distinguish uncertainty, precision, and accuracy. The released spectra do not include an independently calibrated thickness reference, repeated physical measurements, instrument calibration records, or a complete uncertainty budget. Consequently, a traceable standard uncertainty for d cannot be computed from the supplied material. We therefore report an *uncertainty structure* rather than an unsupported numerical accuracy claim.

From Eq. (3), $d \propto (\Delta s)^{-1}$ when the optical factor is fixed. To first order,

$$\frac{\Delta d}{d} \approx -\frac{\Delta(\Delta s)}{\Delta s} + \frac{\partial \ln G^{-1}}{\partial n} \Delta n + \frac{\partial \ln G^{-1}}{\partial \theta_0} \Delta \theta_0. \quad (6)$$

The fringe-spacing sensitivity is therefore exactly $\partial \ln d / \partial \ln(\Delta s) = -1$: a 1% relative spacing error produces an approximately 1% thickness error of opposite sign. For the optical factor used here, the normalized refractive-index sensitivity is

$$S_n = \frac{\partial \ln d}{\partial \ln n} = \frac{n^2[-2n^2 + 3 \sin^2 \theta_0]}{(n^2 - \sin^2 \theta_0)(2n^2 - \sin^2 \theta_0)}, \quad (7)$$

while the angular sensitivity per radian is

$$S_\theta = \frac{\partial \ln d}{\partial \theta_0} = -\frac{\sin^3 \theta_0 \cos \theta_0}{(n^2 - \sin^2 \theta_0)(2n^2 - \sin^2 \theta_0)}. \quad (8)$$

At $n = 2.5$, these expressions give the local sensitivities in Table II. The result makes the error hierarchy transparent: around the present operating point, relative uncertainty in n transfers almost one-for-one to d , whereas small incidence-angle errors have a much weaker first-order effect.

TABLE II
LOCAL SENSITIVITIES OF THE TWO-BEAM ESTIMATOR EVALUATED AT $n = 2.5$. S_n IS DIMENSIONLESS AND S_θ IS PER RADIAN.

θ_0	$G(n, \theta_0)$	S_n	$ S_\theta \text{ (rad}^{-1}\text{)}$
10°	2.500007	-0.999988	6.65×10^{-5}
15°	2.500036	-0.999942	2.18×10^{-4}

The main identifiable contributions are therefore: (i) extrema-location and phase-order uncertainty in Δs ; (ii) systematic error in the dispersion relation used for n ; (iii) incidence-angle uncertainty; (iv) residual polarization effects; (v) absorption and complex-index effects ignored by the high-wavenumber approximation; and (vi) model discrepancy between the two-beam and multiple-reflection descriptions. At the angles used

here, the polarization check suggests that item (iv) is small. The cross-model changes of 0.59% and 0.35% provide an empirical indication that repeated reflections are also a small correction for the two SiC spectra under this analysis.

For a traceable implementation, these terms would enter an explicit uncertainty budget. Spectral-resolution and peak-localization contributions would be evaluated as Type A or Type B components depending on the acquisition protocol; uncertainty in dispersion parameters, angle calibration, and optical-model form would be propagated through the measurement equation. Correlations among inputs would also need to be retained when common calibration information affects multiple extrema. The available competition spectra do not contain the repeated acquisitions or calibration records needed for such a numerical budget, so the present sensitivity coefficients are reported only as analytical diagnostics consistent with the GUM framework [5].

The main limitation is absolute validation. Agreement across two incidence angles and two optical models increases confidence that the phase-spacing calculation is internally stable, but it cannot establish traceability or absolute accuracy. A metrological follow-up should measure the same specimens with an independent reference technique, record instrument spectral resolution and angular uncertainty, propagate uncertainty in the complex refractive index, and compare extrema-based estimation with a full-spectrum transfer-matrix or phase-resolved inversion. Such approaches can explicitly model dispersion and absorption rather than excluding the most distorted spectral region [4].

A second limitation is data provenance at the measurement level. The spectra are public competition data rather than measurements collected by the authors, so the acquisition instrument, specimen history, and raw calibration chain are not under author control. The present conclusions are intentionally restricted to what can be supported by the supplied reflectance series.

VI. MEASUREMENT EQUATION, TRACEABILITY, AND REPRODUCIBILITY

A. From phase spacing to geometric thickness

The local spacing estimator can be derived without fitting an entire reflectance spectrum. For a layer of refractive index n and geometric thickness d , the round-trip optical path inside the layer is $2nd \cos \theta_t$. In wavenumber $s = 1/\lambda$, the propagation phase associated with one round trip is

$$\phi(s) = 4\pi s d n \cos \theta_t. \quad (9)$$

If n varies slowly over a single fringe interval, neighboring extrema correspond approximately to a phase increment of 2π . Hence

$$\Delta\phi \approx 4\pi d \Delta s n \cos \theta_t = 2\pi, \quad (10)$$

which gives the familiar local estimate

$$d \approx \frac{1}{2\Delta s n \cos \theta_t}. \quad (11)$$

The form used in this work is algebraically equivalent after the geometry and interface quantities in the original derivation are

collected into $G(n, \theta_0)$. Snell's law gives $\sin \theta_t = \sin \theta_0/n$, so the angular correction remains close to unity at the modest incidence angles used here. Equation (11) also makes the unit conversion transparent. If Δs is expressed in cm^{-1} , then $1/\Delta s$ is in centimeters and must be converted to micrometers only after the optical factor has been applied.

The approximation behind Eq. (11) is local rather than global. It does not require a constant refractive index over the full 2500–4000 cm^{-1} window; it requires the change in n over one adjacent-extremum interval to be small enough that a single local value is meaningful. The retained calculation indicates approximately 120–135 cm^{-1} fringe spacing in the selected window and a neighboring-fringe refractive-index change around one percent or less. This is why the analysis evaluates the dispersive relation throughout the band rather than replacing it with one constant value for the whole spectrum.

A missed fringe has a qualitatively different effect from a small peak-location error. If two adjacent extrema are mistakenly treated as being separated by two fringe orders, the inferred spacing roughly doubles and the thickness roughly halves. Conversely, inserting a false intermediate extremum produces an overestimate. For that reason, peak-order continuity is a more fundamental validity check than the eventual linear-fit correlation. The first and last reliable extrema anchor the order assignment in the retained workflow, while intermediate locations are regularized by the near-periodic structure. A future implementation should preserve the raw peak list and the assigned orders so that this step can be audited directly.

B. What an uncertainty budget would need to contain

The available spectra support sensitivity analysis, but not a numerical GUM uncertainty statement. A traceable result would require standard uncertainties for the spectral scale, peak localization, refractive-index model, angle setting, repeatability, and model discrepancy. Table III separates these contributions according to how they would normally be established in a laboratory implementation.

If the inputs are collected in a vector $\mathbf{x}$ and the thickness model is $d = f(\mathbf{x})$, the combined standard uncertainty would be formed from the sensitivity coefficients $c_i = \partial f / \partial x_i$ and the covariance matrix of the inputs,

$$u_c^2(d) = \sum_i c_i^2 u^2(x_i) + 2 \sum_{i < j} c_i c_j u(x_i, x_j). \quad (12)$$

The covariance term matters because the same dispersion parameters can affect every fringe and because one spectral calibration can shift multiple peak positions together. Treating all extrema as statistically independent would therefore overstate the amount of independent information in the spectrum. Likewise, the standard deviation of local thickness estimates across extrema is not by itself a complete uncertainty estimate; it measures scatter under a shared model and shared calibration.

For the present data, the most defensible empirical model-discrepancy indicator is the change between the two-beam and Airy-type estimates: 0.59% for the 10° spectrum and 0.35% for the 15° spectrum. Those values can be reported as observed cross-model differences. They should not be

TABLE III
MEASUREMENT INPUTS THAT WOULD BE REQUIRED FOR A TRACEABLE UNCERTAINTY BUDGET. THE PRESENT DATASET DOES NOT CONTAIN ENOUGH CALIBRATION INFORMATION TO ASSIGN ALL OF THESE STANDARD UNCERTAINTIES NUMERICALLY.

Quantity	Typical source	Evaluation route	Relevance to the present estimator
Spectral scale / wavenumber	FTIR calibration and spectral resolution	Type B from calibration certificate; Type A if repeated reference scans are available	Directly affects extrema positions and therefore Δs
Peak localization	detector noise, resolution, smoothing, peak-selection rule	Type A from repeated spectra or resampling; Type B from the peak-picking algorithm	Transfers nearly one-for-one to thickness through $d \propto (\Delta s)^{-1}$
Refractive index $n(s)$	dispersion parameters and material state	parameter covariance, literature uncertainty, or independent ellipsometric/IR calibration	Local normalized sensitivity is approximately -1 at the operating point
Incidence angle θ_0	mechanical alignment and beam divergence	calibrated stage specification and repeated alignment	Small first-order effect at 10° – 15° , but can grow at larger angle
Polarization	source/optics polarization state	measured polarization ratio or bounded Fresnel calculation	Small for the unpolarized average in the present geometry
Model discrepancy	omitted absorption, roughness, incoherence, multilayer structure	comparison with transfer-matrix/phase-resolved model and reference samples	Not captured by fit correlation; cross-model comparison provides only a partial diagnostic
Repeatability	repeated placement and acquisition	Type A standard deviation under a fixed protocol	Not available from the single released spectrum for each condition
Reference thickness	certified step, cross-section, or independent calibrated method	comparison to a traceable reference	Required for an accuracy/traceability statement; absent from the released data

inserted mechanically as a complete Type B uncertainty because both models share the same spectra, dispersion relation, peak locations, and much of the same phase structure. Their errors are correlated.

C. Model validity before numerical uncertainty

Uncertainty propagation is meaningful only after the measurement equation is judged adequate for the data. Three checks are especially relevant. First, the usable spectral window must contain identifiable, coherent fringes; the strong lattice-vibration region is excluded here because the regular pattern is visibly disrupted. Second, the layer should be sufficiently uniform over the illuminated spot that a single thickness is a reasonable measurand. A thickness gradient would broaden or chirp the fringes and could make a single-spacing estimate misleading. Third, the optical model must retain the physical effects that are large enough to change phase or fringe order. Strong absorption, rough interfaces, additional layers, or large-angle polarization effects would require a complex-index transfer-matrix or phase-resolved treatment rather than the present local estimator.

These checks place the correlation values in context. A transformed spectrum can be highly periodic even if every fringe is shifted by the same incorrect refractive-index model. Such a case would produce excellent internal fit consistency and a biased geometric thickness. Conversely, a slightly noisier spectrum could have a lower correlation while still giving an unbiased estimate once the peak positions and calibration are handled correctly. The fit diagnostic is useful for identifying gross inconsistency, but it is not a substitute for model validation or reference comparison.

D. Reproducibility record for a future laboratory implementation

A practical replication package should preserve more than the final thickness. At minimum it should contain the raw re-

flectance spectrum, background/reference spectrum, wavenumber calibration information, incidence angle and alignment record, polarization state, spectral window used for fitting, all detected extrema with their assigned orders, the refractive-index relation and parameter source, the exact optical model, and the script that converts those inputs to thickness. If smoothing or baseline correction is applied, the unprocessed spectrum and all filter parameters should remain available.

The same record should separate observations from analyst choices. The measured reflectance and instrument settings are observations. The decision to exclude 800 – 1000 cm^{-1} , the choice of extrema, and the decision to treat the layer with a two-beam or Airy-type model are analysis choices. Preserving both makes later disagreement diagnosable: a reviewer can tell whether two estimates differ because the spectra differ, because a fringe was assigned differently, or because the optical model changed.

For the released competition data, not all of this provenance exists. The present paper therefore keeps the reproducible claim at the level actually supported by the archive: the reported spectra, the stated window, the documented formulas, and the retained numerical thickness results can be inspected; a full metrological traceability chain cannot be reconstructed.

VII. MODEL COMPARISON AND SCOPE OF THE RESULT

The two optical treatments answer slightly different questions. The two-beam model asks whether the dominant fringe period can be mapped to thickness with a simple, auditable phase relation. The Airy-type model asks whether retaining coherent repeated reflections materially changes that recovered thickness. Their close agreement for the SiC spectra says that this particular model refinement is small in the selected window. It does not show that every omitted effect is small.

This hierarchy is also why the silicon results are not used as an accuracy standard for the SiC estimates. They show that the multiple-reflection calculation can be applied to spectra with

TABLE IV
INTERPRETATION OF THE TWO OPTICAL MODELS USED IN THIS WORK.

Model	Physics retained	Main advantage	Main limitation
Dispersive two-beam	dominant surface and substrate-reflected components, local dispersion, incidence-angle correction	direct relation between fringe spacing and thickness; easy sensitivity analysis	simplified amplitude response; repeated reflections and complex absorption are not modeled explicitly
Airy-type multiple reflection	coherent geometric series of internal round trips in addition to the same phase-spacing structure	checks whether repeated reflection changes the recovered thickness	still not a full complex-index, roughness-aware, multilayer inversion
Transfer-matrix / phase-resolved model (not fitted here)	complex refractive index, absorption, multilayers, full amplitude and phase treatment	most complete route when those effects are important	requires more optical parameters, stronger data provenance, and a larger inverse problem

a different refractive-index regime and return finite thickness values, but the silicon wafers do not provide certified SiC reference thicknesses. The result of this paper is therefore deliberately limited: within the archived spectra and selected high-wavenumber window, the SiC thickness estimate is stable across two angles and across one explicit increase in optical-model complexity.

VIII. POLARIZATION AND INTERFACE-REFLECTANCE CHECK

The scalar treatment used above assumes that the unpolarized response can be represented adequately by one effective interface reflectance at the small incidence angles in the dataset. This can be checked directly from the Fresnel coefficients. For incidence from air, with refractive indices $n_0 \approx 1$ and n , the power reflectances for s and p polarization are

$$R_s = \left| \frac{n_0 \cos \theta_0 - n \cos \theta_t}{n_0 \cos \theta_0 + n \cos \theta_t} \right|^2, \quad (13)$$

$$R_p = \left| \frac{n \cos \theta_0 - n_0 \cos \theta_t}{n \cos \theta_0 + n_0 \cos \theta_t} \right|^2. \quad (14)$$

For an unpolarized beam the first-order interface reflectance is approximated by $R_u = (R_s + R_p)/2$. At $\theta_0 = 15^\circ$ and $n \approx 2.5$, the retained calculation gives $R_s \approx 19.41\%$, $R_p \approx 17.35\%$, and $R_u \approx 18.38\%$. The corresponding normal-incidence scalar value is about 18.37%. The near equality of these two averages is why polarization splitting is not carried as a dominant correction in the present thickness estimates.

This check has a limited purpose. It does not imply that s and p components are individually identical, nor does it justify ignoring polarization at arbitrary angle. Near Brewster conditions, or in anisotropic layers for which the optical response depends on crystal orientation, the two components can differ materially. A polarization-resolved experiment would then require separate transfer functions and, potentially, tensor optical constants. The present conclusion is only that the unpolarized average at 10° – 15° is close enough to the scalar interface value that polarization is unlikely to explain the observed 0.35–0.59% cross-model changes.

The same logic applies to interface reflectance in the Airy-type calculation. Repeated internal reflections become influential when the round-trip amplitude remains large enough to survive several passes through the layer. That depends jointly on interface reflectance, absorption, surface parallelism,

and coherence. A low-loss layer with appreciable reflectance can show a pronounced Airy response; strong absorption or roughness suppresses later round trips even if the first interface reflectance is large. Thus, “multiple reflection” is not a binary material property. It is a condition of the complete optical path and spectral region. The selected high-wavenumber window is useful partly because the regular fringe pattern indicates that a coherent phase relation survives there.

A. Why cross-angle agreement is a diagnostic rather than a calibration

The two SiC spectra correspond to different external angles but nominally the same type of epitaxial structure. The two-beam estimates differ by $0.10\mu\text{m}$, or about 1.18% relative to the $8.50\mu\text{m}$ value. This is reassuring because the angle enters the optical path explicitly; a gross error in the angular correction would tend to separate the estimates. Nevertheless, cross-angle agreement cannot determine which estimate is closer to the true geometric thickness. Both calculations use the same refractive-index relation, similar fringe-selection logic, and spectra produced by the same unknown measurement chain. Shared systematic error can move both values in the same direction.

For the same reason, averaging the two angle results would not create a traceable reference value. A future experiment could make the angle comparison quantitatively stronger by measuring one certified sample repeatedly at several calibrated angles and by fitting all spectra with a shared thickness parameter. The residual angle dependence would then be a direct diagnostic of model inadequacy or alignment error. The current archive does not support that joint fit, so the two-angle result remains an internal check only.

IX. CONCLUSION

The four released FTIR spectra support a useful consistency test of fringe-based thickness estimation, but not an absolute calibration. For the two SiC cases, the dispersive two-beam model gives $8.50\mu\text{m}$ and $8.60\mu\text{m}$, while the Airy-type treatment gives $8.55\mu\text{m}$ and $8.63\mu\text{m}$. The small model-to-model changes, together with the modest cross-angle difference, show that the recovered thickness is stable to these two modeling choices in the selected spectral window. The sensitivity calculation also makes clear where a real uncertainty budget should focus: fringe spacing and refractive index matter much more than small angle errors at 10° – 15° .

What the data do not provide is equally important. There is no certified reference thickness, no instrument calibration record, and no repeated acquisition series from which a traceable standard uncertainty can be built. The numerical agreement reported here should therefore be read as internal consistency. A metrological implementation would need an independent reference, calibrated spectral and angular scales, repeated measurements, and a fuller optical model where absorption or multilayer effects are material.

DATA AVAILABILITY

The reflectance spectra analyzed in this work were released as attachments to the 2025 China Undergraduate Mathematical Contest in Modeling (CUMCM), Problem B. The manuscript reports author-generated plots and derived quantities from those spectra but does not redistribute the original competition attachments. Readers seeking the underlying spectra should obtain them from an official or organizer-authorized source and comply with the organizer's distribution terms.

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